GlucoTwin AI - End-to-End System Architecture

Multimodal Physiological Digital Twin for Type 2 Diabetes Management

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1. High-Level Data Ingestion & Microservices Topology

The architecture couples asynchronous IoT wearable feeds with patient Electronic Health Records (EHR) into a unified inference engine.

2. Subsystem Functional Breakdown

• **Client Layer (Frontend SPA):** React 19, Vite, Recharts, and Tailwind CSS v4 running glassmorphic dashboards with live parameter sliders and WhatsApp API dispatch.

• **Inference Layer (FastAPI):** RESTful routing executing PyTorch forward passes and Bergman Minimal Model kinetic calculations in <15ms.

• **Model Framework:** Dual-branch neural network optimizing joint Mean Squared Error (trajectory regression) and Cross-Entropy loss (clinical alert classification).

• **Security & Governance:** Stateless zero-storage inference aligned with Indian DPDP Act 2023 and HIPAA synthetic validation protocols.